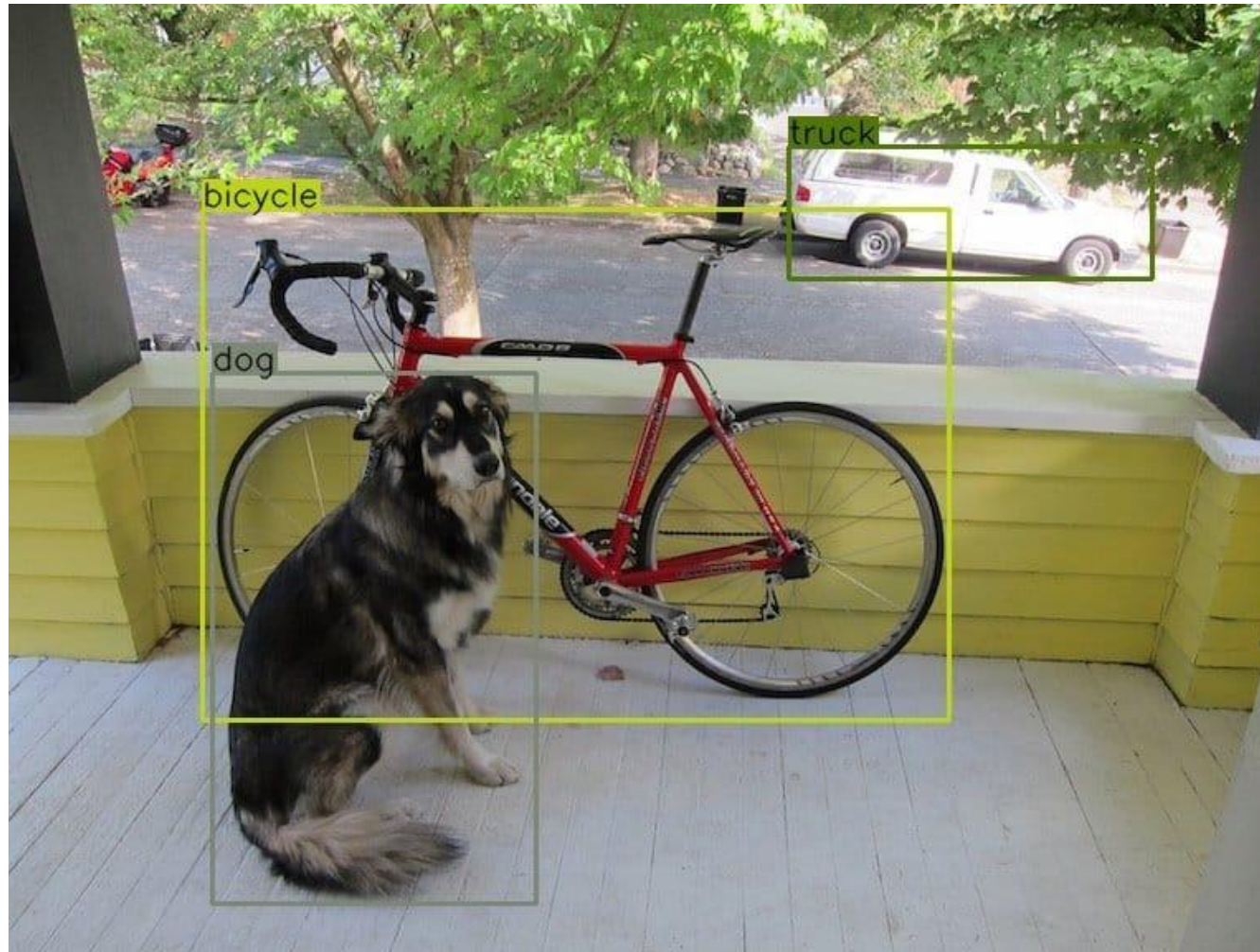


Computer Vision II

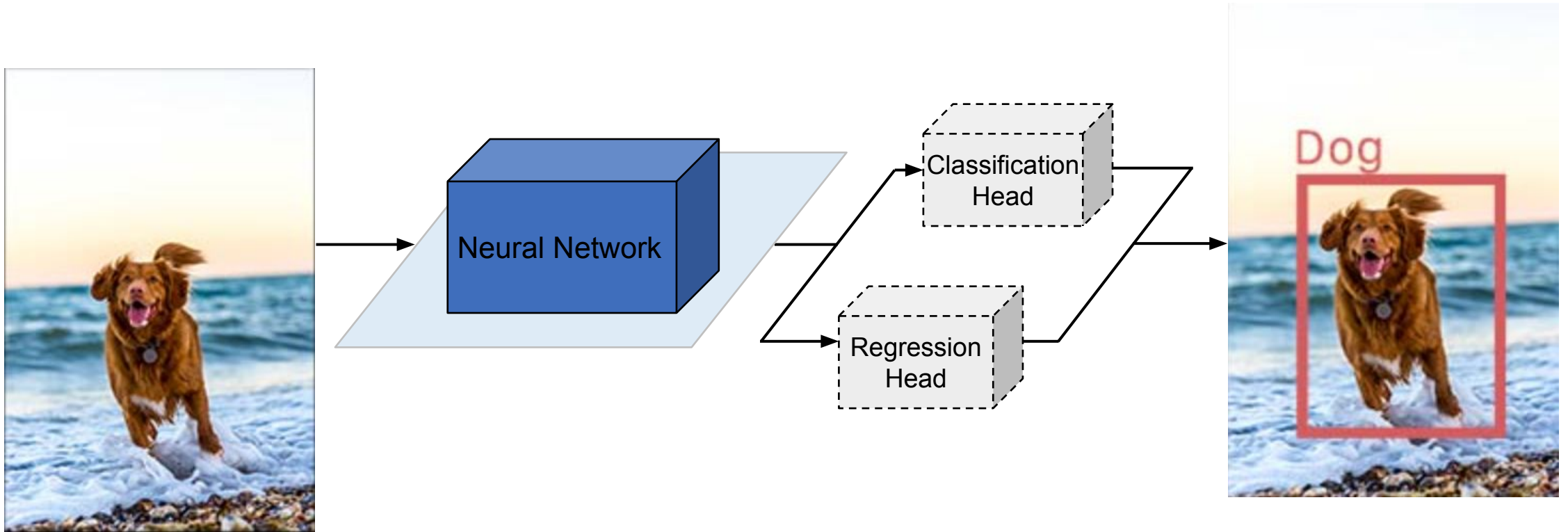
Guosheng Hu

bristol.ac.uk

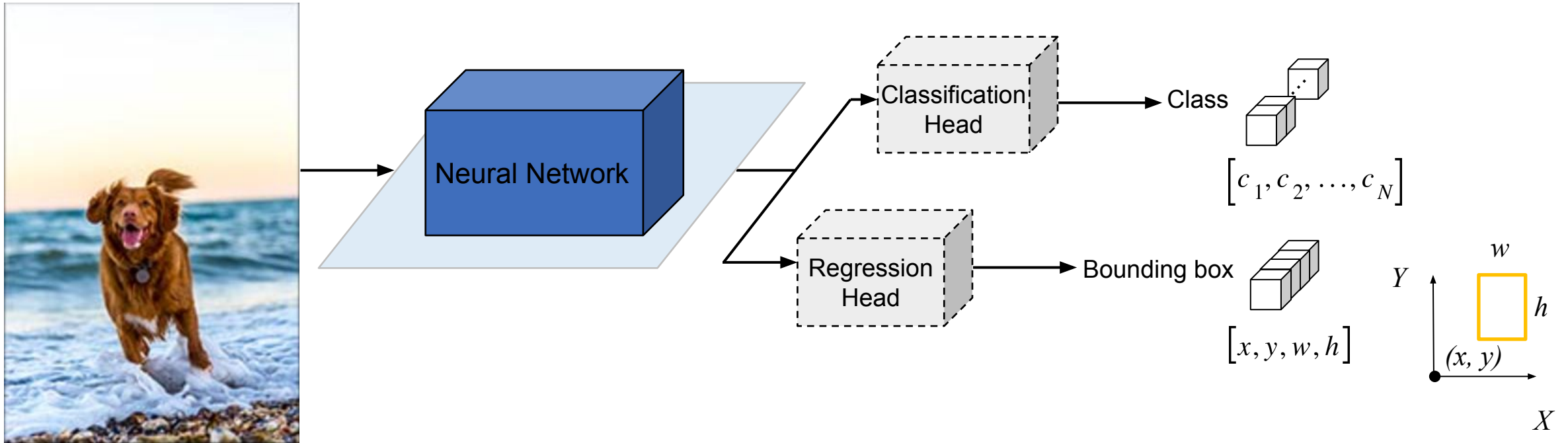
Object Detection (What and Where)



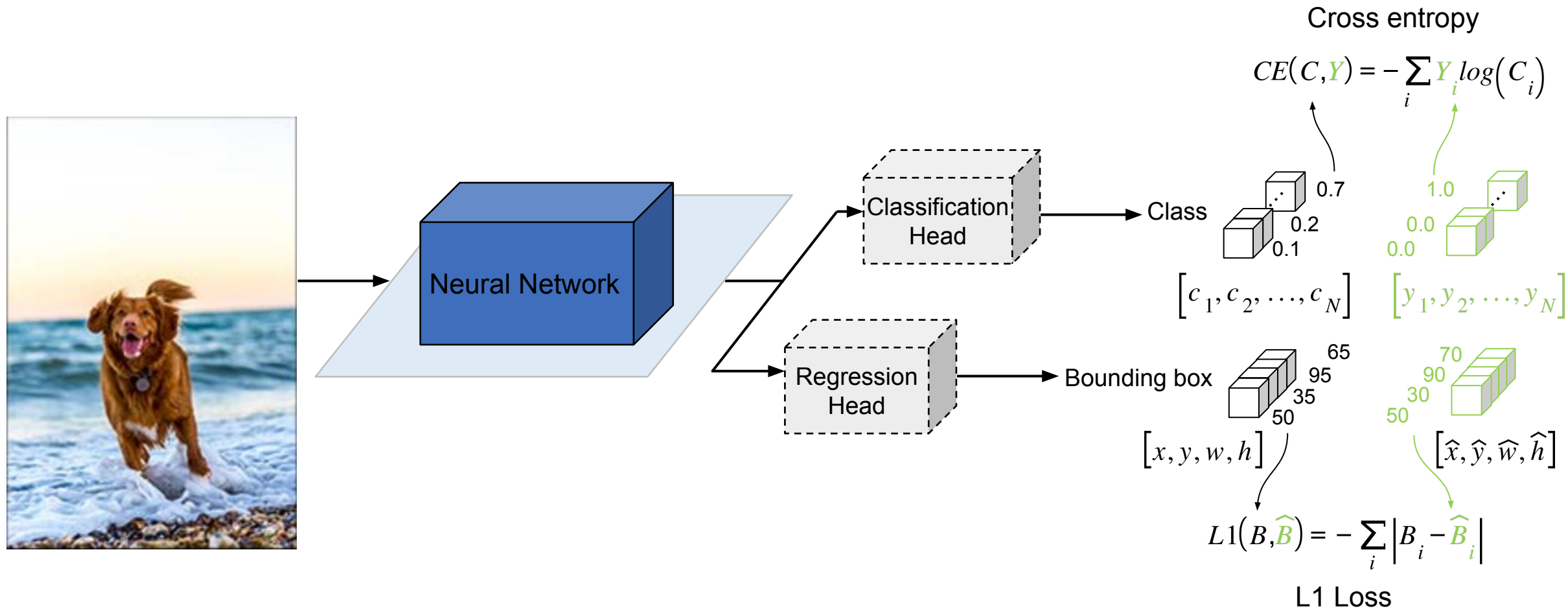
Object Detection (Intuition)



Object Detection (One Prediction)

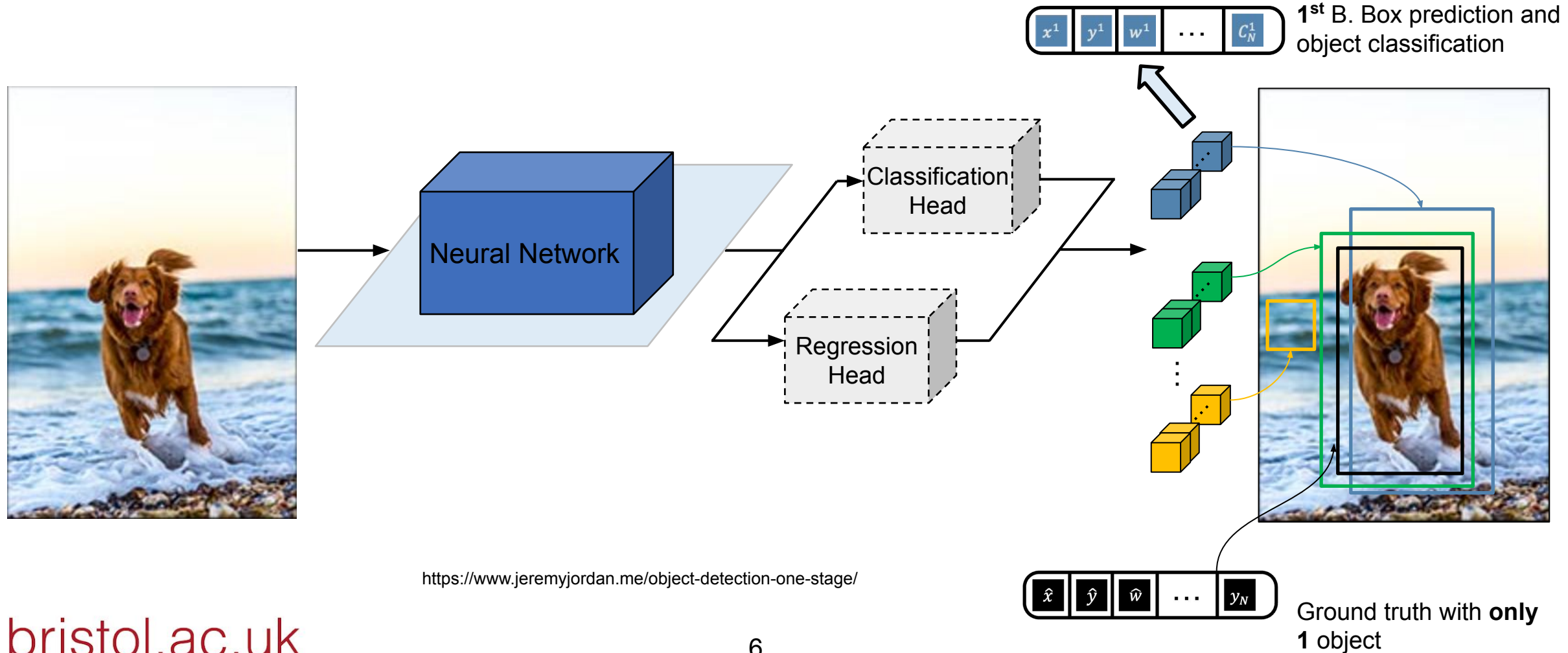


Object Detection (Loss Function)

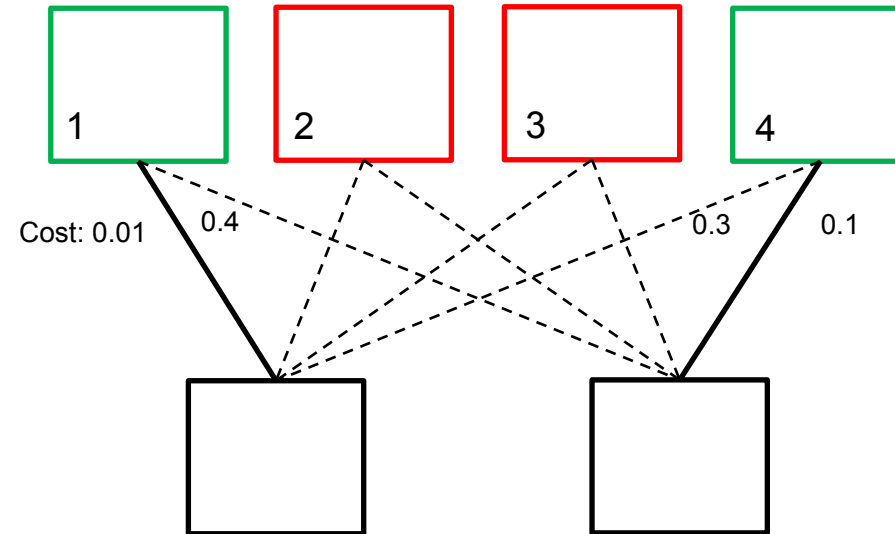
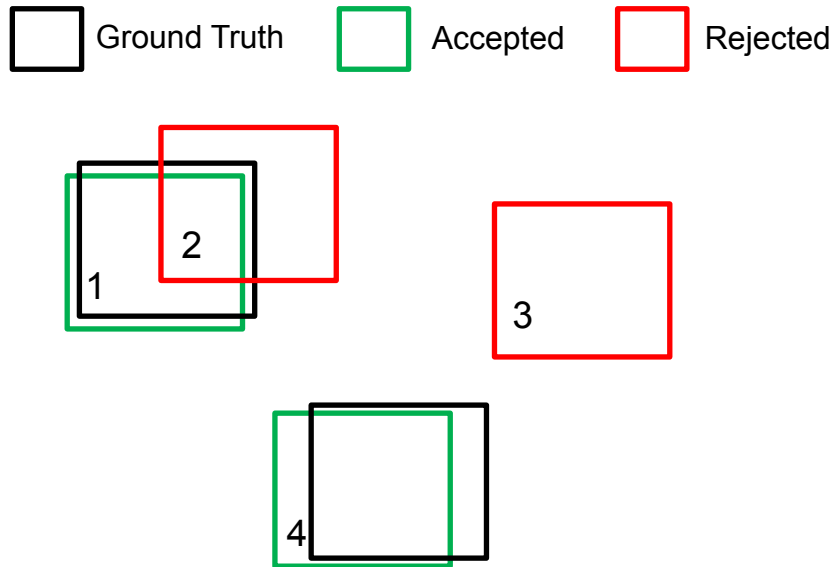


***K* Predictions vs. 1 Object (GT)**

In fact, the network will perform a **predefined number of predictions (K)** to handle single or multiple objects, or no object



Hungarian Matching (K predictions and M objects)



Stewart et al. "End-to-end people detection in crowded scenes." *CVPR*, 2016.

Hungarian Matching (K predictions and M objects)

For a $n \times n$ cost matrix, Hungarian matching involves 4 steps:

1. Row Reduction. Subtract the minimum value of each row from each element in that row.
 2. Column Reduction. Subtract the minimum value of each column from each element in that column.
 3. Cover the zeros with the fewest number of straight lines. If the number of lines equals n (3 in example), the optimal solution is found. Otherwise, go to step 4.
 4. Find the smallest value δ (2 in example) among the uncovered elements. Subtract δ from all uncovered numbers. **Add δ to numbers where two lines intersect.** Then, return to step 3.
 5. Results: Company C has only one zero (Musician). A has Musician and cleaner, but musician is chosen by C, so cleaner for A. B can only choose Chef.
- Total cost: $150 + 135 + 122 = \$407$.

<https://brilliant.org/wiki/hungarian-matching>

Company	Cost for Musician	Cost for Chef	Cost for Cleaners
Company A	\$108	\$125	\$150
Company B	\$150	\$135	\$175
Company C	\$122	\$148	\$250

			Step 1			Step 2		
108	125	150	0	17	42	0	17	2
150	135	175	15	0	40	15	0	0
122	148	250	0	26	128	0	26	88
Step 3			Step 4			Step 3		
0	17	2	0	15	0	0	15	0
15	0	0	17	0	0	17	0	0
0	26	88	0	24	86	0	24	86

The assignment is made where each row and column has exactly one 0

Hungarian Matching (4 predictions and 2 objects)

Smaller value means better match.

	p1	p2	p3	p4
GT1	0.2	0.8	0.5	0.9
GT2	0.6	0.1	0.9	0.7
	0	0	0	0
	0	0	0	0

Step 1			
0.0	0.6	0.3	0.7
0.5	0.0	0.8	0.6
0	0	0	0
0	0	0	0

Step 2			
0.0	0.6	0.3	0.7
0.5	0.0	0.8	0.6
0	0	0	0
0	0	0	0

Step 3			
0.0	0.6	0.3	0.7
0.5	0.0	0.8	0.6
0	0	0	0
0	0	0	0

4 lines cover all zeros. Done.
p1 -> GT1, p2->GT2

Commonly Used Loss Function

As mentioned before, to handle single or multiple objects, the neural network will perform $\mathbf{K}$ predictions. $\mathbf{K}$ is larger than the total number of classes N (including one extra class: no object $\emptyset$) and the number of objects exist in the each image in dataset.

We define the following matching cost for the potential subset of predictions σ and perform Hungarian matching to find the optimal subset $\hat{\sigma}$:

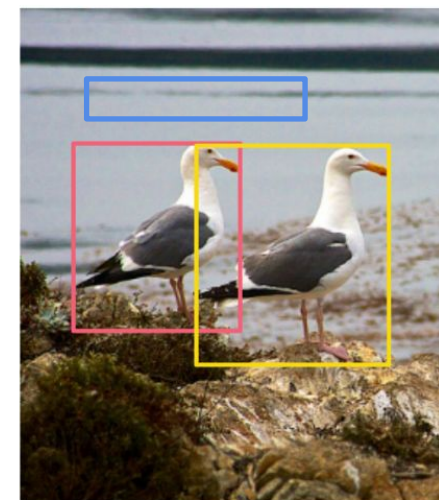
$$-\mathbb{1}_{\{c_i \neq \emptyset\}} c_{\sigma(i)}(y_i) + \mathbb{1}_{\{c_i \neq \emptyset\}} L_{box}(B_{\sigma(i)}, \hat{B}_i)$$

For the matched subset $\hat{\sigma}$ with minimum cost, we could then adjust the weights to optimize the following objective loss function:

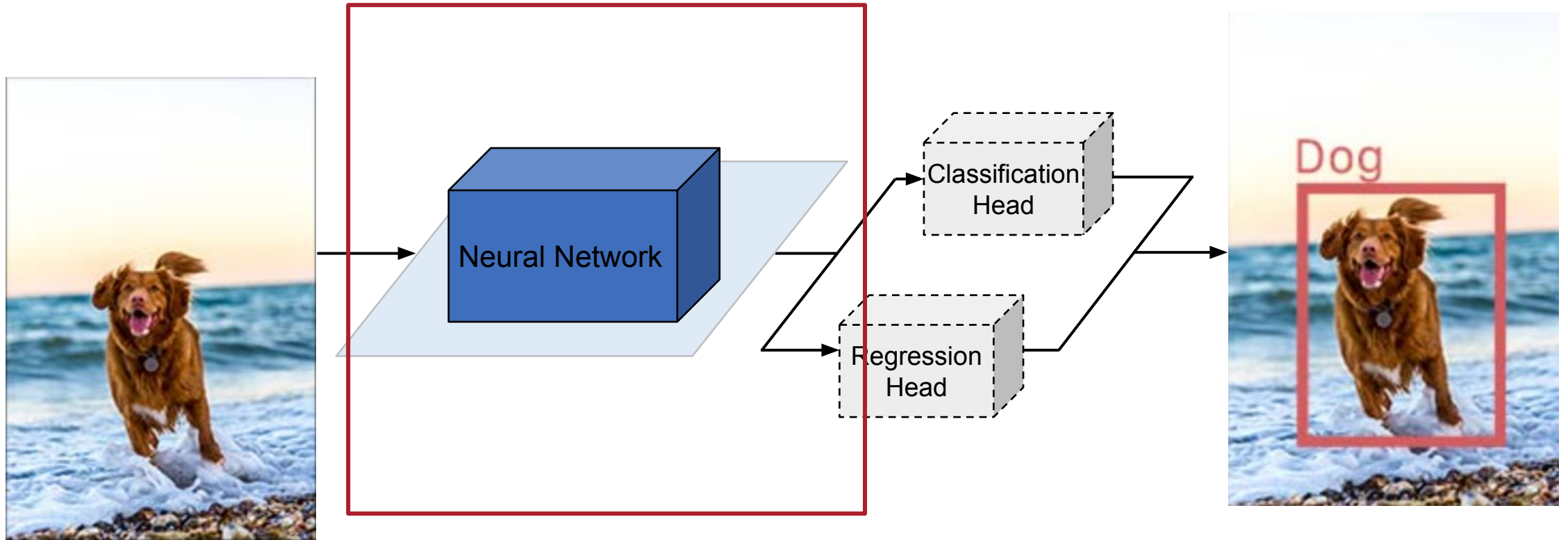
$$L_{box} = \lambda_{iou} L_{iou}(B_i, \hat{B}_{\hat{\sigma}}(i)) + \lambda_{L1} L1(B_i, \hat{B}_{\hat{\sigma}}(i))$$

$$L_{iou}(B_i, \hat{B}_{\hat{\sigma}}(i)) = 1 - \frac{\text{Area}(B_i \cap \hat{B}_{\hat{\sigma}}(i))}{\text{Area}(B_i) + \text{Area}(\hat{B}_{\hat{\sigma}}(i)) - \text{Area}(B_i \cap \hat{B}_{\hat{\sigma}}(i))}$$

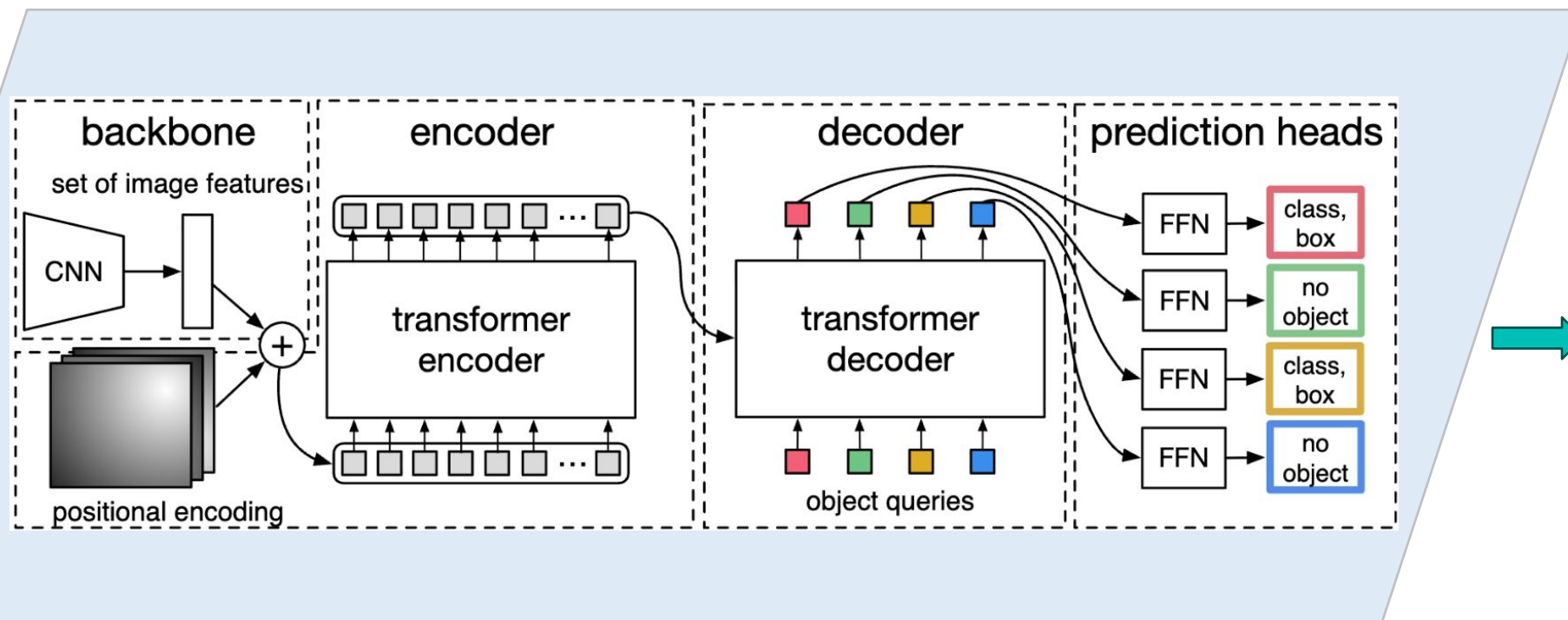
For instance, $\mathbf{K}=3$, $N=2$



What is this Neural Network?



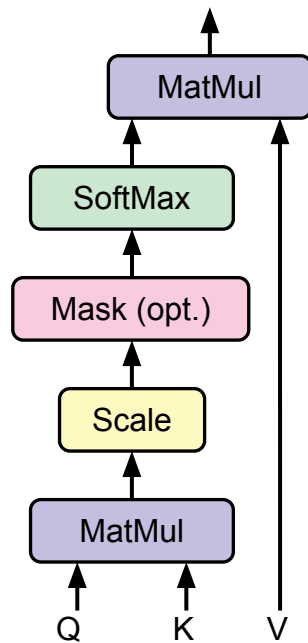
DEtection TRansformer (DETR)



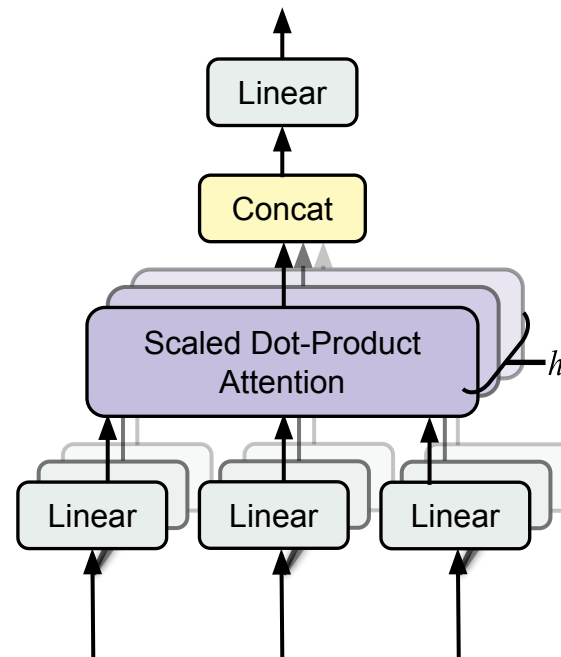
Carion, Nicolas, et al. "End-to-end object detection with transformers." *ECCV*, 2020.

Inspired by Transformer

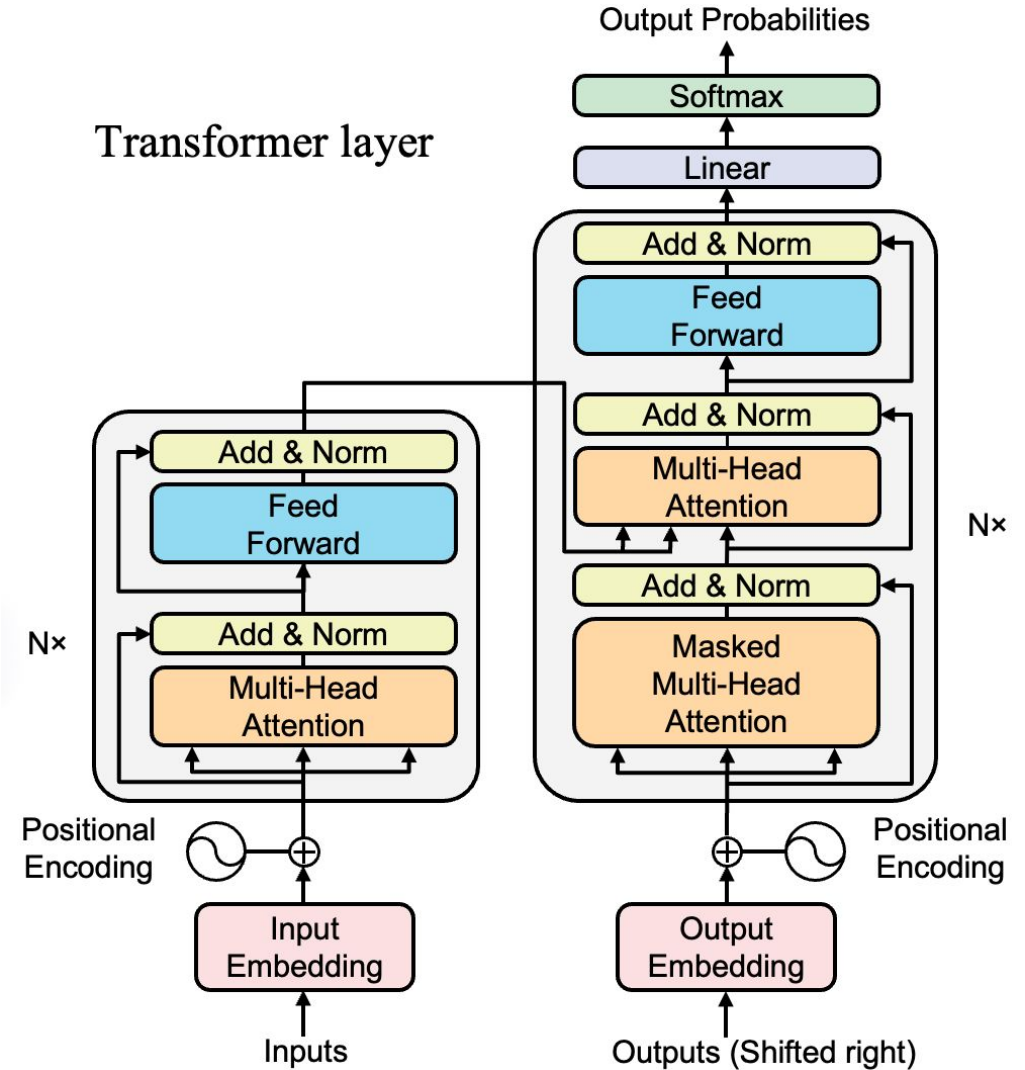
Scaled Dot-Product Attention



Multi-Head Attention



Transformer layer

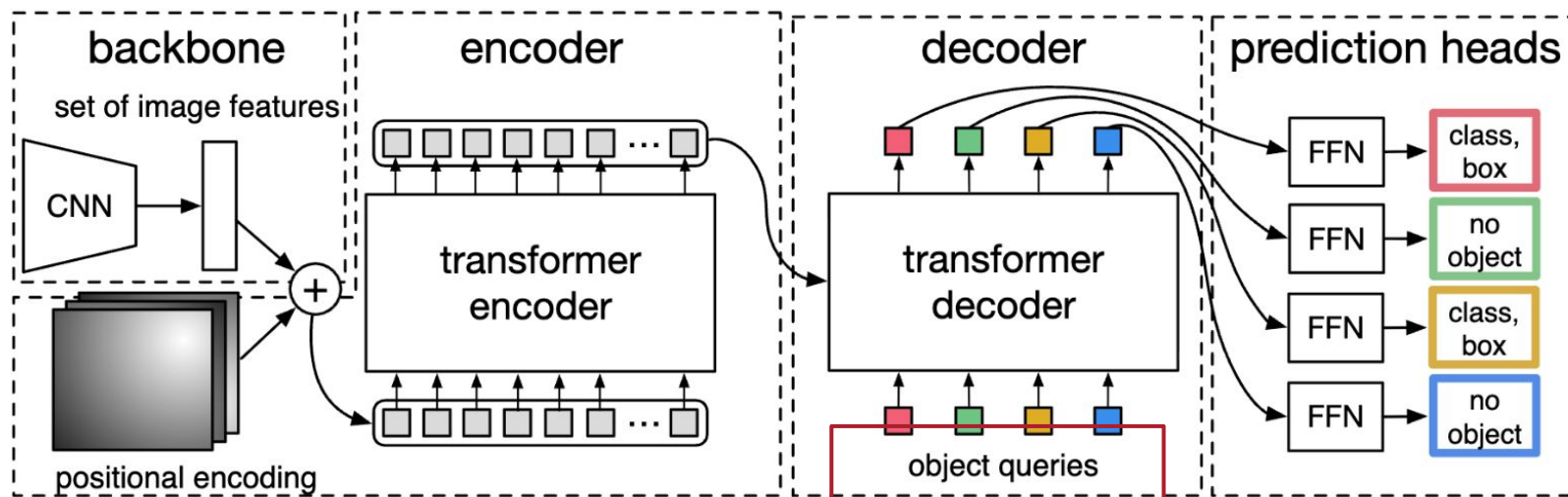


Vaswani, Ashish, et al. "Attention is all you need." *NeurIPS*, 2017.

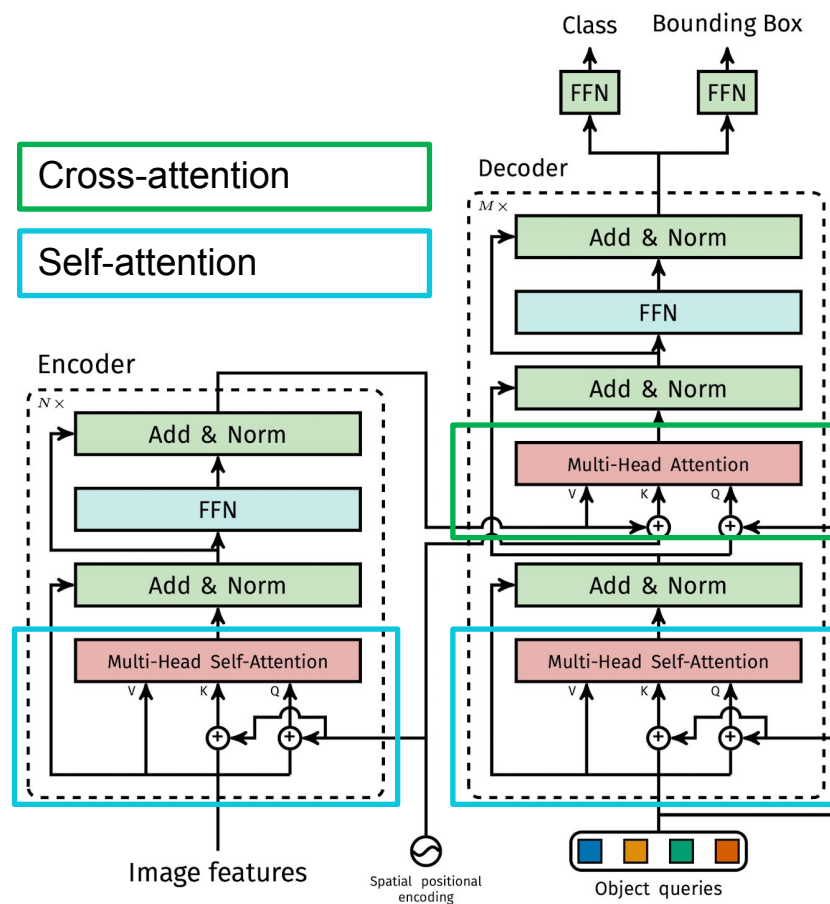
Object Queries

Learnable parameters, like an intermediate layer between encoder and decoder.

- Queries act as **object slots**: each one focuses on one potential object.
- They let the **decoder attend selectively** to parts of the encoder feature map.
- They provide a **fixed-size, parallelizable output** for set prediction.

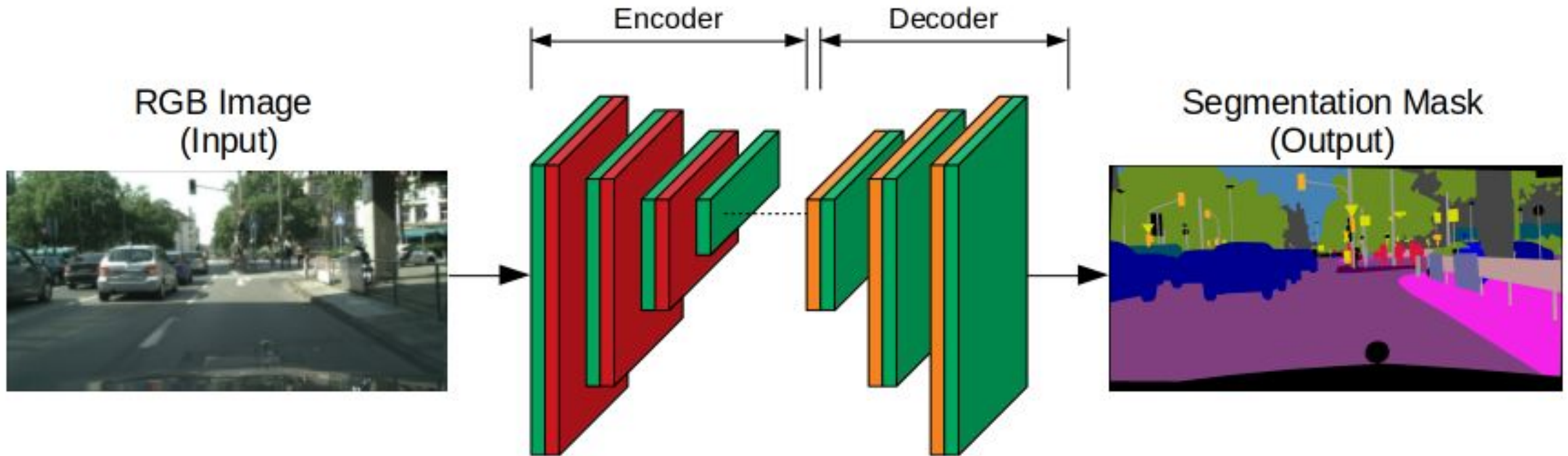


Object Queries - Cross Attention



Carion, Nicolas, et al. "End-to-end object detection with transformers." *ECCV*, 2020.

Semantic Segmentation

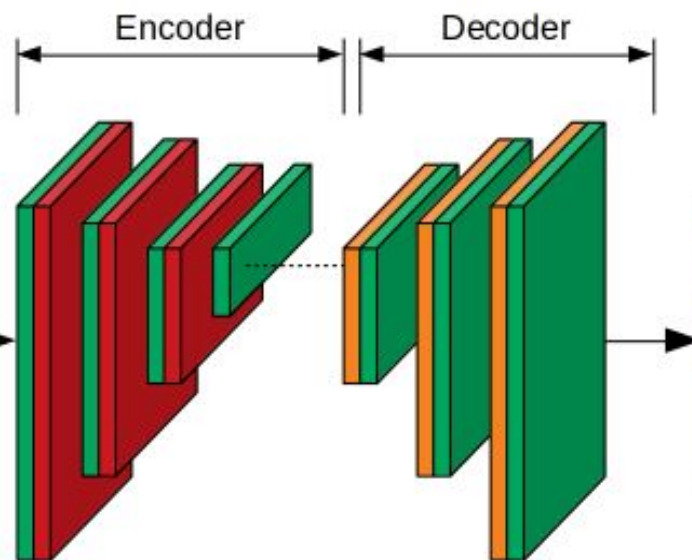


Noh, Hyeonwoo, et al. "Learning deconvolution network for semantic segmentation." /CCV, 2015.

Pixel-Level Classification

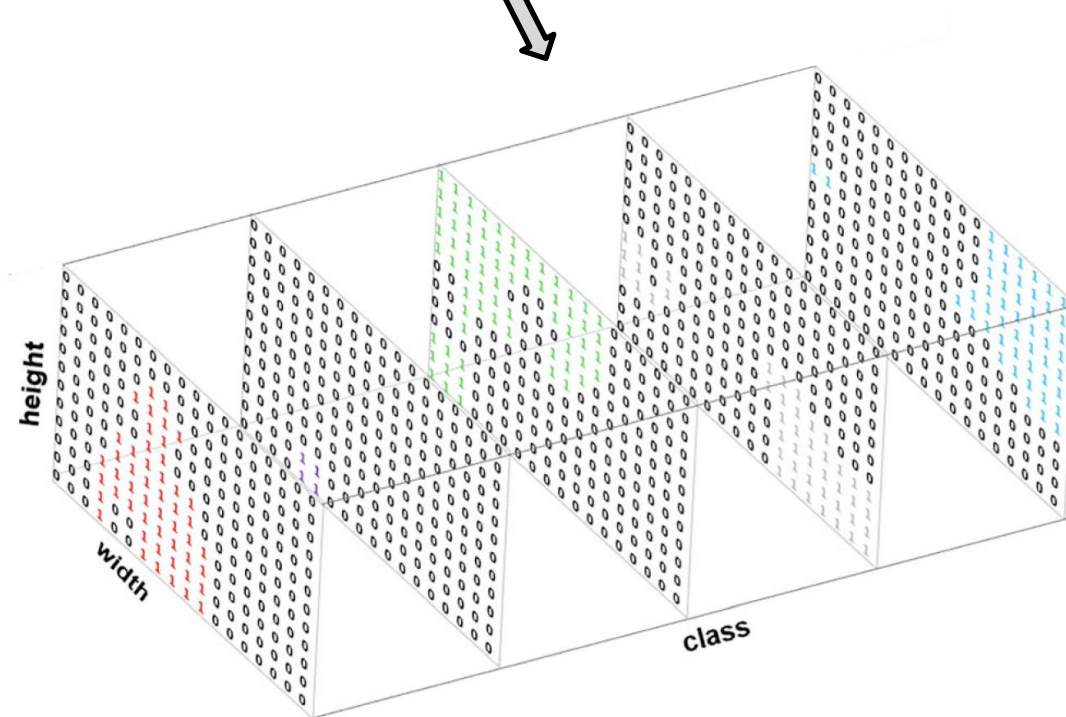


RGB Image
(Input)



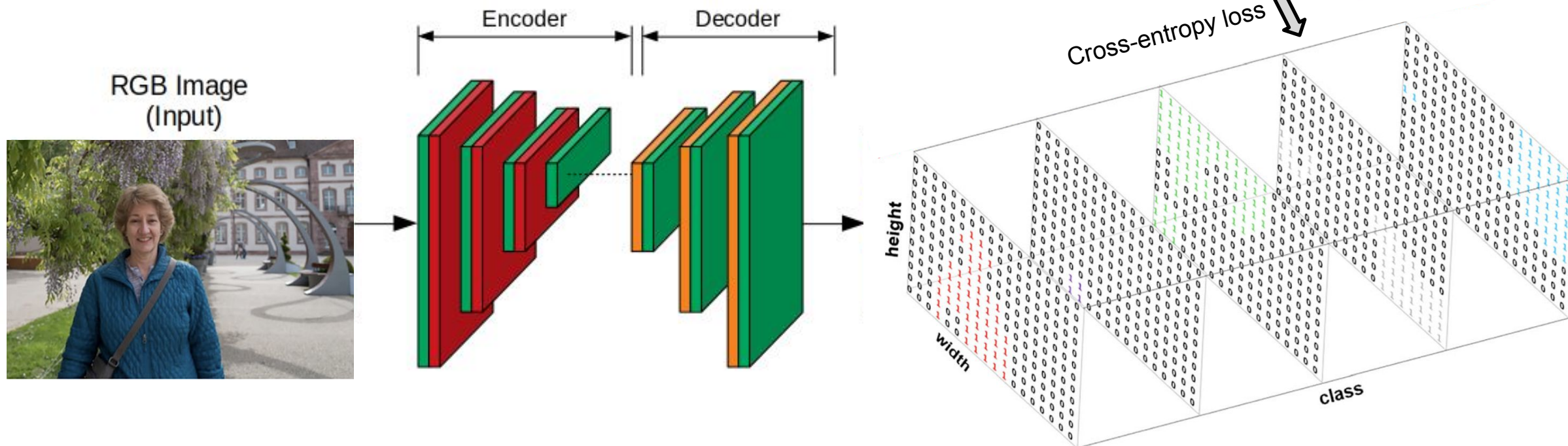
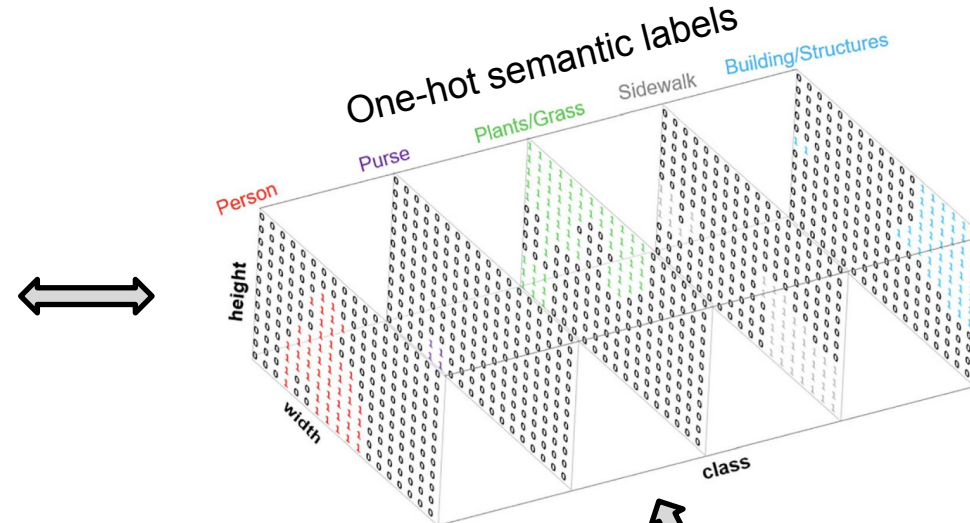
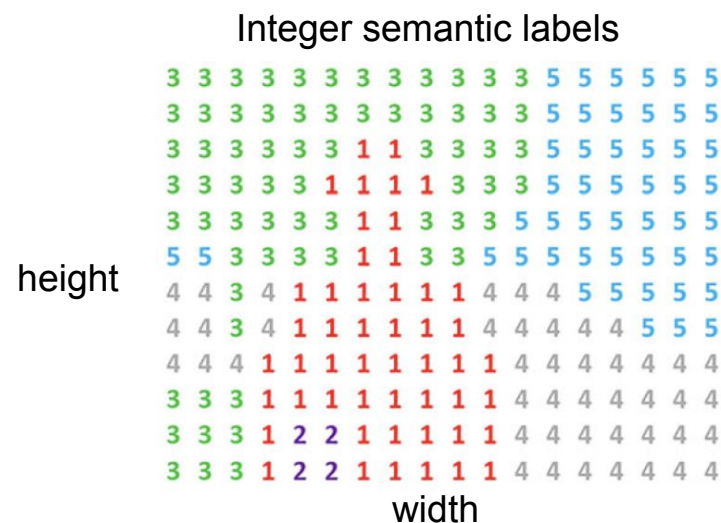
height

width



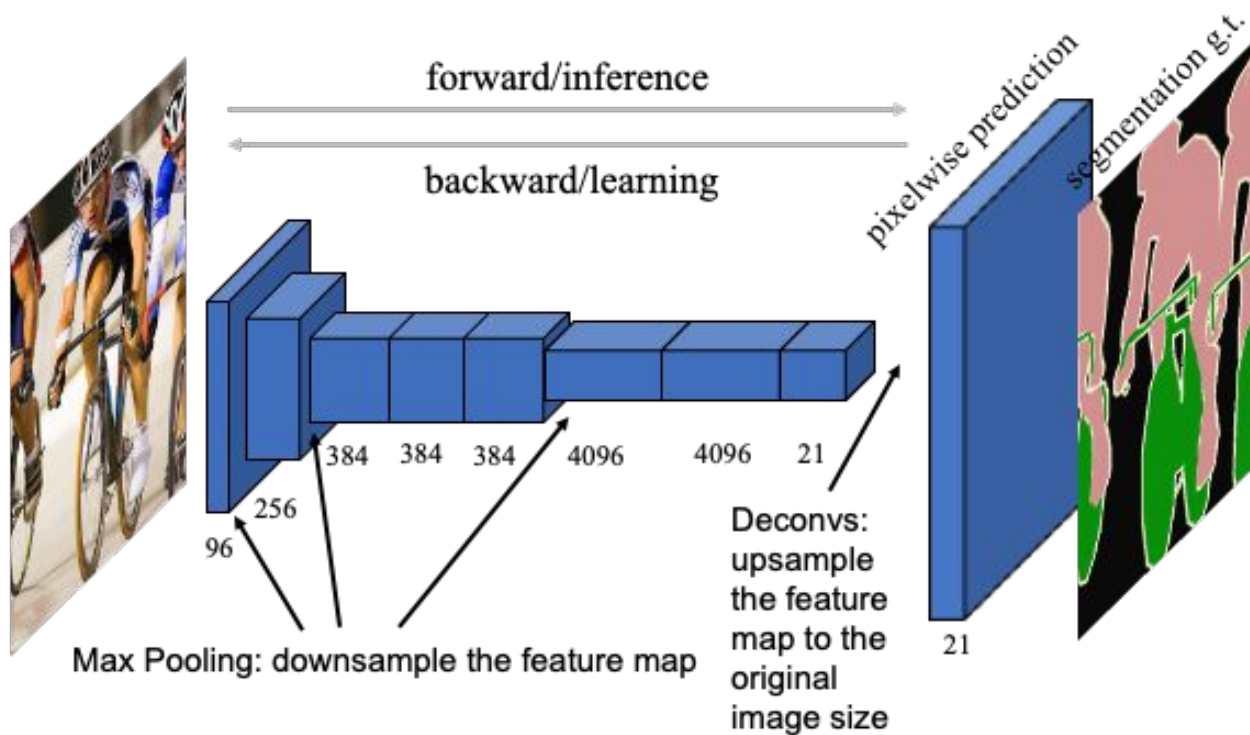
Gao, Yan, et al. "SFSM: sensitive feature selection module for image semantic segmentation." *Multimedia Tools and Applications*, 2023

Label



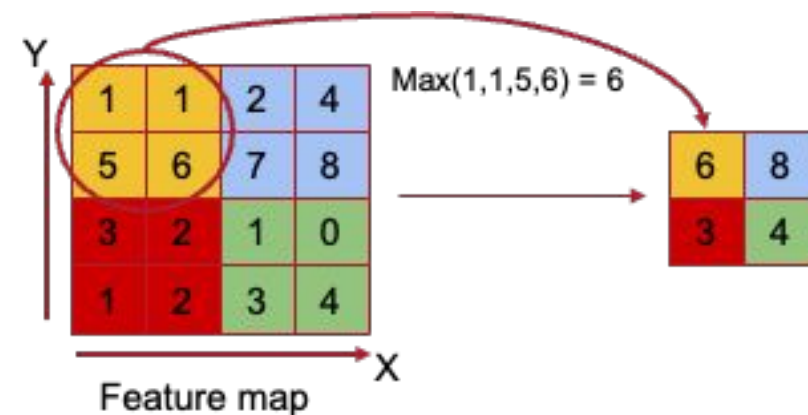
Long, Jonathan, et al. "Fully convolutional networks for semantic segmentation." *CVPR*, 2015.

Network Design - FCN

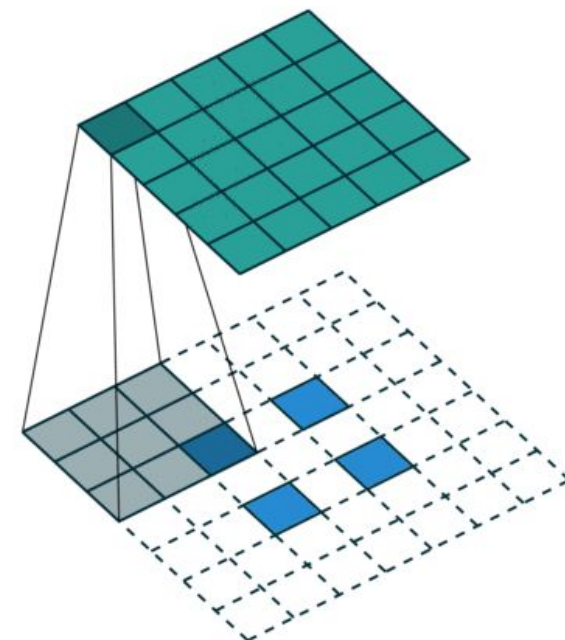


Long, Jonathan, et al. "Fully convolutional networks for semantic segmentation." *CVPR*, 2015.
 Dumoulin et al. "A guide to convolution arithmetic for deep learning." *arXiv*, 2016.

Max Pooling with 2x2 kernel and stride of 2



Deconvolution (3x3 kernel, stride of 2)



The resolution of the output (5x5) is higher than the one of the input (2x2)

Deconvolution/Transposed Conv

Calculations are same as normal convolution, but with different stride and padding meaning, and the size of output is larger than the input.

Considering padding p , kernel size k , stride s :

Step 1. Padding:

Fill $s-1$ rows and columns zeros (or specified values) between the input feature map elements.

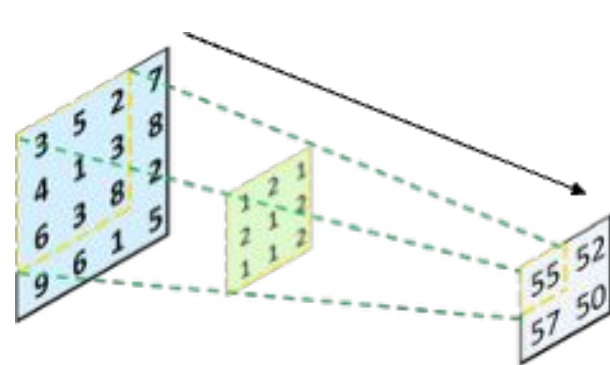
Fill $k-p-1$ rows and columns zeros (or specified values) around the input feature map.

Step 2. Sliding the kernel:

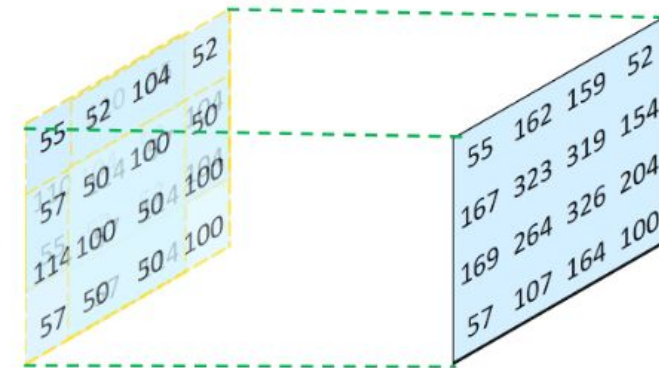
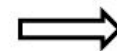
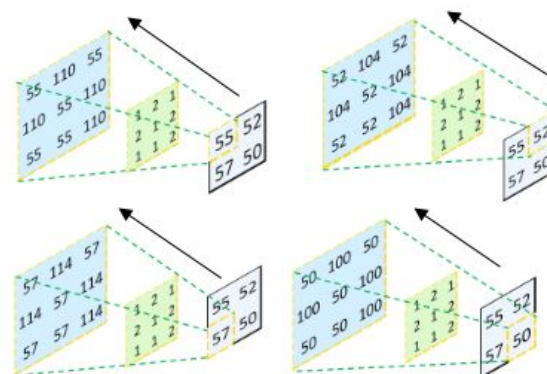
In **deconvolution**, the kernel always move **only with a stride length of 1 on the input**. The stride s and padding p are the parameters of original convolution.

Thus when stride s is larger and padding p is smaller, the output size of deconvolution will be larger.

Normal convolution

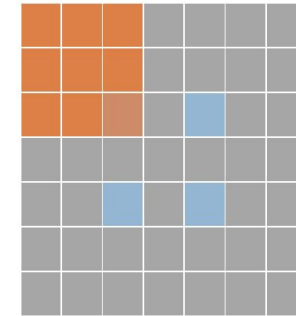


Deconvolution/Transposed Conv ($s=1, p=0$)

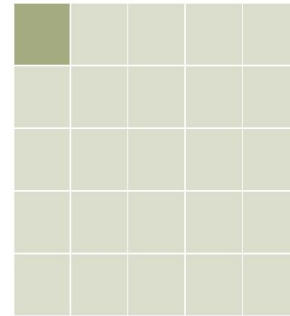


Combine all four resulting layers with corresponding positions and sum the overlapped values

Type: transposed'conv - Stride: 2 Padding: 0

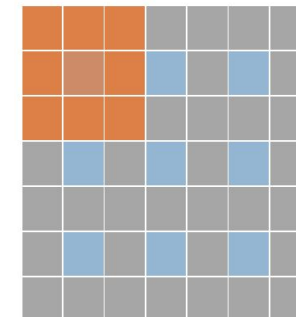


Input

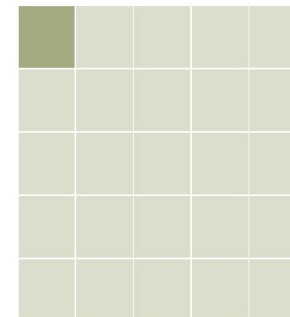


Output

Type: transposed'conv - Stride: 2 Padding: 1

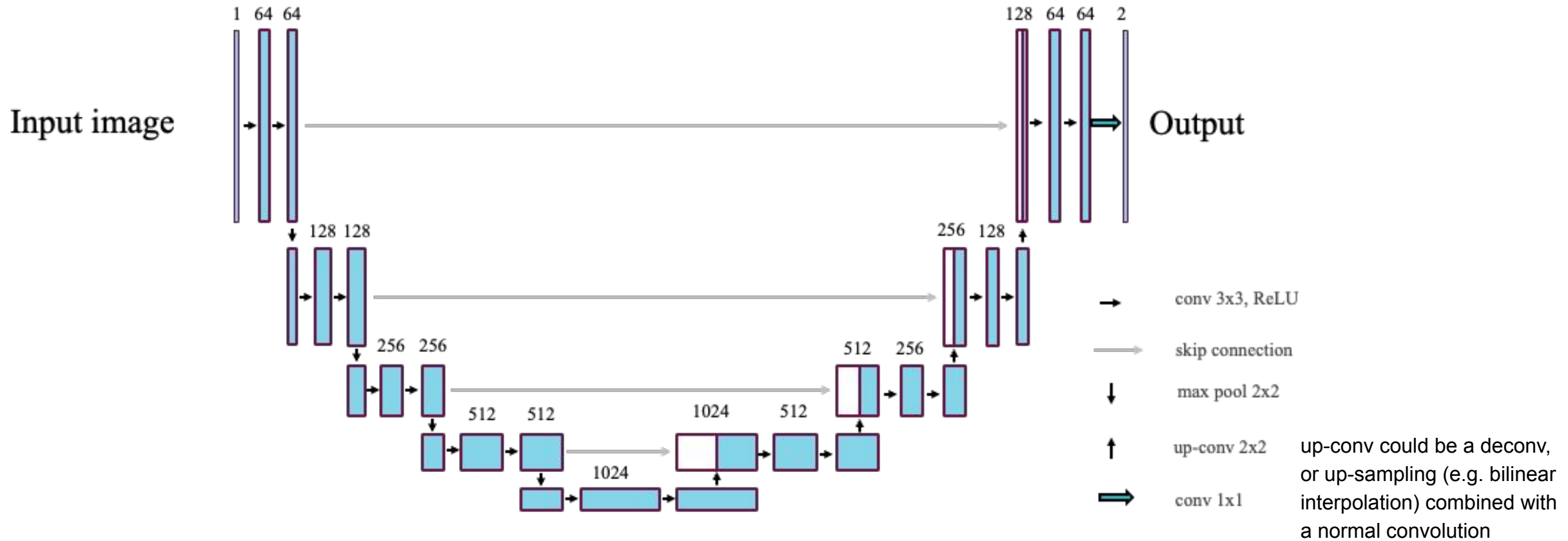


Input



Output

Network Design - U-Net



Ronneberger, Olaf, et al. "U-net: Convolutional networks for biomedical image segmentation." *MICCAI*, 2015.

Reading List

Girshick, Ross. "Fast r-cnn." *Proceedings of the IEEE international conference on computer vision*. 2015.

Ren, Shaoqing, et al. "Faster R-CNN: Towards real-time object detection with region proposal networks." *IEEE transactions on pattern analysis and machine intelligence* 39.6 (2016): 1137-1149.

Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.

Tian, Zhi, et al. "Fcos: Fully convolutional one-stage object detection." *Proceedings of the IEEE/CVF international conference on computer vision*. 2019.

Liu, Wei, et al. "Ssd: Single shot multibox detector." *European conference on computer vision*. Cham: Springer International Publishing, 2016.

Kuhn, Harold W. "The Hungarian method for the assignment problem." *Naval research logistics quarterly* 2.1-2 (1955): 83-97.